Sagnac Effect Entry Report

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1 Q1

We're assuming the two plane wave beams are linearly polarized in the same direction and have the same frequency, i.e the waves are

$$\mathbf{E_1} = E_1 \cos\left(\mathbf{k_1}\cdot\mathbf{r} - \omega t + \phi_1\right) \tag{1}$$
$$\mathbf{E_2} = E_2 \cos\left(\mathbf{k_2}\cdot\mathbf{r} - \omega t + \phi_2\right) \tag{2}$$

so as is written in the manual the intensity distribution is

$$I = \left|\mathbf{E_1} + \mathbf{E_2}\right|^2 = \left|\mathbf{E_1}\right|^2 + \left|\mathbf{E_2}\right|^2 + 2\left|\mathbf{E_1}\cdot\mathbf{E_2}\right|^2 \tag{3}$$
$$= I_1 + I_2 + 2\sqrt{I_1 I_2}\cos\left(\left(\mathbf{k_1} - \mathbf{k_2}\right)\cdot\mathbf{r} + \Delta\phi\right)\left(\hat{e_1}\cdot\hat{e_2}\right) \tag{4}$$

next we know that there is an angle of θ between the beams' direction of movement, both have same amplitude wavevector, and since there are two beams we could always change the base vectors to have them described by only two, we can choose arbitrarily to call it the y axis

$$\mathbf{k_1} - \mathbf{k_2} = 2ky\sin\theta \tag{5}$$

giving us

$$I = I_1 + I_2 + 2\sqrt{I_1 I_2}\cos\left(2ky\sin\theta + \Delta\phi\right)\cos\alpha \tag{6}$$

where θ is the angle between the directions of the beams' movement, ϕ is the phase offset between them, and α is the angle between their polarization direction.

2 Q2

If we insert $k = 2\pi/\lambda$ and look only at the last element of eq. 1, ignoring the polarization offset,

$$\cos\left(\frac{4\pi y}{\lambda}\sin\theta + \Delta\phi\right) = 1 \tag{7}$$

then we instead solve

$$\frac{4\pi y}{\lambda}\sin\theta = 2\pi n - \Delta\phi \tag{8}$$

simplifying to

$$y = \frac{\lambda}{2\sin\theta}\left(n - \frac{\Delta\phi}{2\pi}\right) \tag{9}$$

meaning the distance between maxima is

$$\Lambda = \frac{\lambda}{2\sin\theta} \tag{10}$$

3 Q3

The amplitude of the electric field of a gaussian beam, assuming polarization in the x direction and movement in the z direction is

$$\mathbf{E} = E_0\hat{x}\frac{w_0}{w(z)}\exp\left(\frac{-r^2}{w(z)^2}\right)\exp\left(-i\left(kz + k\frac{r^2}{2R(z) - \psi(z)}\right)\right) \tag{11}$$

where r is the radial distance from the center axis of the beam, z is the axial distance from beam's waist, i is the imaginary unit, k is the wavenumber, E_0 is the amplitude at the origin, $w(z)$ is the radius at which the amplitude falls to $1/e$ of the axial value, w_0 is the waist radius, $R(z)$ is the radius of curvature of the beam's wavefronts at z, and $\psi(z)$ is the Gouy phase at z.

4 Q4

As we saw the travel times in the different directions are

$$t_+ = \frac{L}{c - v} \tag{12}$$
$$t_- = \frac{L}{c + v} \tag{13}$$

so

$$\Delta t = \frac{2Lv}{c^2 - v^2} \tag{14}$$

approximating for speeds far below c

$$\Delta t = \frac{2Lv}{c^2} \tag{15}$$

inserting $L = 2\pi R$ and $v = \omega R$

$$\Delta t = \frac{4\pi\omega R^2}{c^2} = \frac{4A\omega}{c^2} \tag{16}$$

where A is the area of the light's travel. The phase offset therefore is just

$$\Delta\phi = \omega_c \Delta t \tag{17}$$

where $\omega_c = \frac{2\pi c}{\lambda}$, so

$$\Delta\phi = \frac{2\pi c}{\lambda}\frac{4A\omega}{c^2} = \frac{8\pi A\omega}{\lambda c} \tag{18}$$

but if our fiber is looped N times the phase offset is just

$$\Delta\phi = N\frac{8\pi A\omega}{\lambda c} \tag{19}$$

5 Q5

What we measure in the experiment is the intensity, so let's take eq. 1 and massage it

$$I = I_1 + I_2 + 2\sqrt{I_1 I_2}\cos\left(2ky\sin\theta + \Delta\phi\right)\cos\alpha \tag{20}$$

inserting the Sagnac formula and setting $\alpha = 0$ for convenience, and looking at non-constant terms

$$\delta I \propto \cos\left(2ky\sin\theta + \Delta\phi\right) \tag{21}$$

some algebra gives

$$\delta I \propto \cos\left(\frac{2\pi y}{\Lambda} + \Delta\phi\right) \tag{22}$$

looking for maxima of the cosine

$$\frac{2\pi y}{\Lambda} + \Delta\phi = 2\pi m \tag{23}$$

so

$$y_m = m\Lambda - \frac{\Lambda\Delta\phi}{2\pi} \tag{24}$$

looking at the phase offset element

$$\Delta y = -\frac{\Lambda\Delta\phi}{2\pi} \tag{25}$$

rearranging to

$$\Delta\phi = -2\pi\frac{\Delta y}{\Lambda} \tag{26}$$

Another method that could be done is to fit a cosine

$$I = C + B\cos\left(2\pi y/\Lambda + \phi\right) \tag{27}$$

to the measurements. We expect the fit method to be more precise for large phase changes, since this method uses all the points measured to improve the result, compared to being calculated for each point separately.

6 Q6

In general, reflecting light preserves its polarization. Using this fact we can use the following method:

1. Set a polarizer at the laser, polarizing the light. Reflections will thus have this polatization.
2. The light passes through the fiber, exiting with a different polarization due to Berry phase.
3. Set a polarizer at the camera set to be perpendicular to the first polarizer.

Malus' law dictates that reflections will be blocked, and if the light exiting the fiber will have a sufficient Berry phase, some of it will pass.

7 Q7

The highlighted values in the schematic in the question direct us towards the gaussian thin lens equation

$$\frac{1}{d} + \frac{1}{s} = \frac{1}{f} \tag{28}$$

where s is the distance at which the lens will focus the rays again. We can see that when $d = f$ the rays coming out from the lens wil be parallel. When $d < f$ they will diverge (since the focal point s will be negative) and for $d > f$ they will converge on a point. This applies both to regular light coming from the fiber and reflected light, since we're assuming it's a point source.

8 Q8

It doesn't, and it would be very unphysical if it did. We can also see this from the Sagnac formula, it depends on the area the light traces out, not the position of the center.

9 Q9

The equation for the distance between maxima in the interference pattern is eq. 10, where we can see it is equal to the wavelength multiplied by a factor depending on the angle. To this the Sagnac effect adds a contribution, but in the opposite direction. The higher the wavelength, the smaller the impact of the Sagnac effect.

10 Q10

It does not, when both the fiber and interferometer rotate together. The relativistic calculation ends up canceling out the effects from refractive index.

11 Q11

From the manual:

wavelength	1550 nm	1310 nm
NA	0.14	0.14
n_{eff}	1.4679	1.4674

and the core diameter is 8.2 µm

12 Q12

From the manual, fibers with $V <\sim 2.405$ are single mode fibers. Inserting the numbers for our fiber into the formula for V:

$$V = \frac{2\pi}{\lambda}a \cdot NA \tag{29}$$

$$= \cdots = 5.635 \tag{30}$$

so our fiber is not a single mode fiber.

13 Q1 from the maual

The optimum from the manual is $W_0 = 3.16\,\mu\text{m}$. Throwing this together with the wavelength into the equation for W_0

$$NA = \frac{2\lambda}{\pi W_0} = \cdots = 0.1289 \tag{31}$$

and throwing everything into the other equation for W_0

$$f = \frac{\pi D W_0}{4\lambda} = \cdots = 15.51\,\text{mm} \tag{32}$$

14 Q2 from the manual

coupling efficiency is given by

$$T_{GB} = \frac{4w_0^2 w_{eff}^2}{\left(w_0^2 + w_{eff}^2\right)^2} \tag{33}$$

w_{eff} is given by the Marcuse approximation

$$\frac{w_0}{a} \approx 0.65 + \frac{1.619}{V^{3/2}} + \frac{2.879}{V^6} \tag{34}$$

plugging numbers gives us

$$w_{eff} = 3.16\,\mu\text{m} \tag{35}$$

and the waist is, using $f = 4.5\,\text{mm}$ and $W = 1\,\text{mm}$

$$w_o' = \frac{\lambda f}{\pi W} = \cdots = 0.92\,\mu\text{m} \tag{36}$$

giving an efficiency of

$$T_{GB} = \cdots = 0.29 \tag{37}$$

to get the best coupling we want a lens with $w_0' \sim w_{eff}$, or

$$f_{best} = \frac{\pi W w_{eff}}{\lambda} = 15.5\,\text{mm} \tag{38}$$

and the closest lens is the RMS10x.